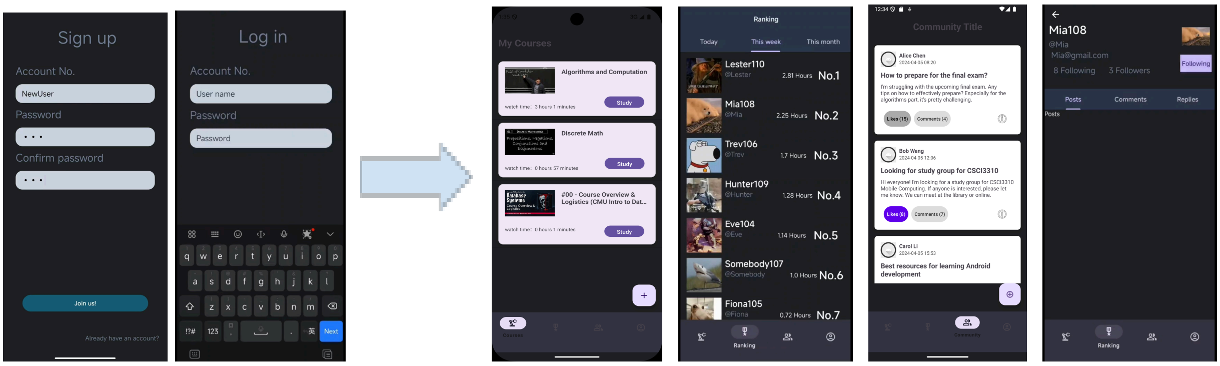
# Phase II Report: LearnLeague

## 1. App Layout Flow

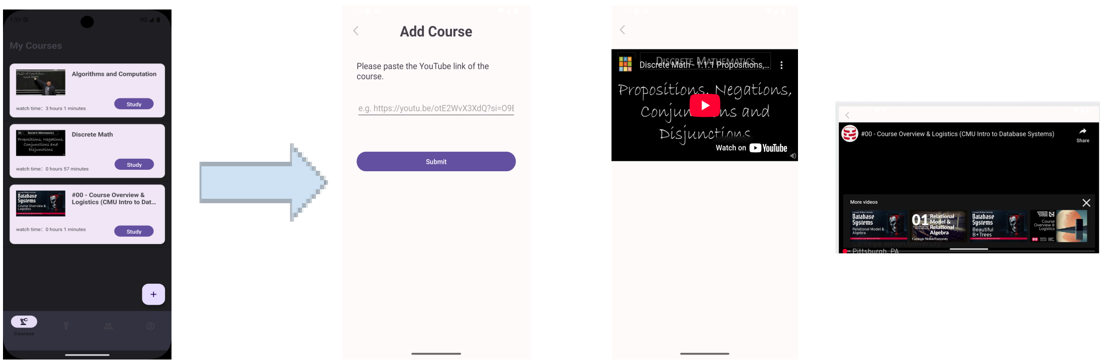
LearnLeague starts with a **Register/Login page**, and then goes to the **Course page** and provides access to our 4 key features via a bottom navigation bar, which are **Course, Ranking, Community and Profile.**




Sign up / Login page | Course | Ranking | Community | Profile

### Course

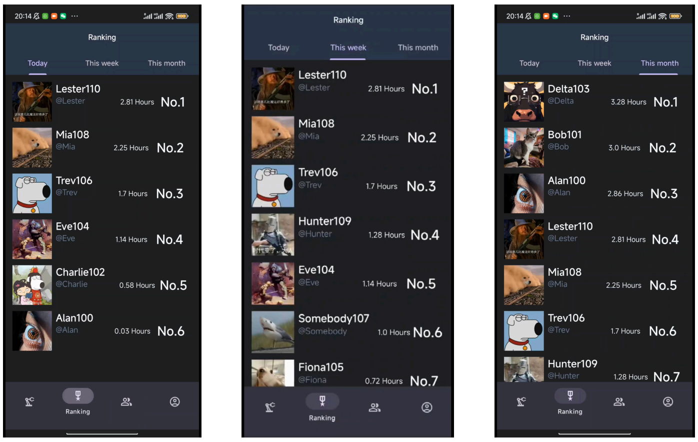
The Course page allows navigation to the Add Course page via the '+' button and to the Video Player page via the 'Study' button. On the Video Player page, users can watch the video in both portrait and landscape mode. There are back < buttons on both the Add Course page and Video Player page for users to go back to the course page.



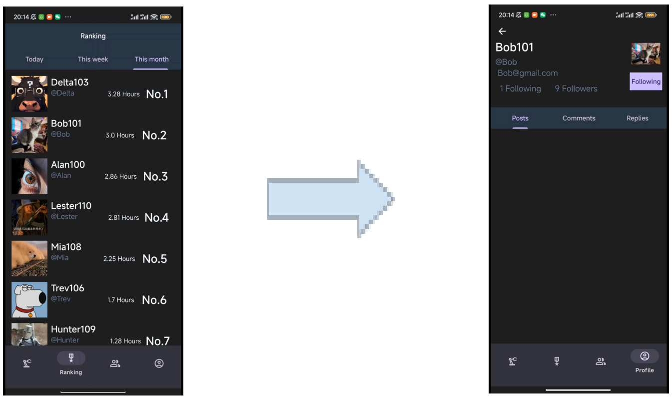

Add Course | Video Player

### Ranking

There are three sub-pages on the ranking page, which are **Today**, **This week**, and **This Month**, users can simply click on the corresponding time period to navigate to the leaderboard according to different periods. Users are also allowed to click on other users' icons to access their profile to know more about them or follow them by simply clicking the **Follow** button.



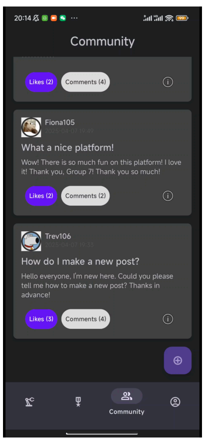

Leaderboards of **Today, This week, and This month**



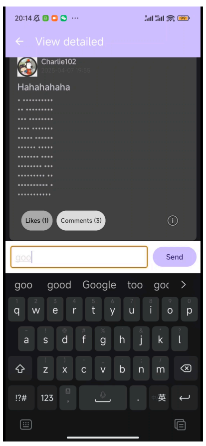

Navigate to other user's profile

**Community**

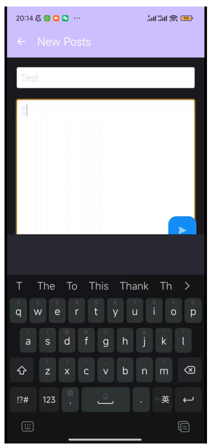
After clicking the community icon, you will be taken to the community page where you can view all posts from other users. On the bottom right of each thread is an exclamation mark icon. By clicking it, you can view the detailed information of the original post. To create a new post, simply click the 'add' icon, you will be redirected to the New Posts page where you can enter a title and detailed content and create a new post. Users can also check out other user's profiles in this community page.




Post Playground

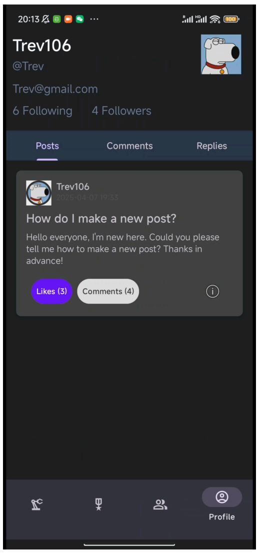



Comment

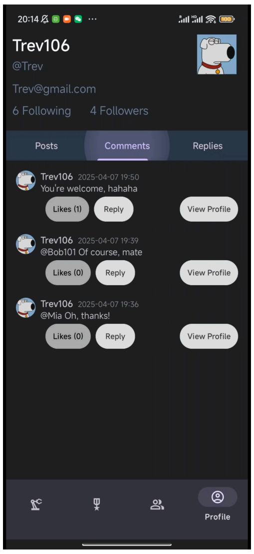



New post

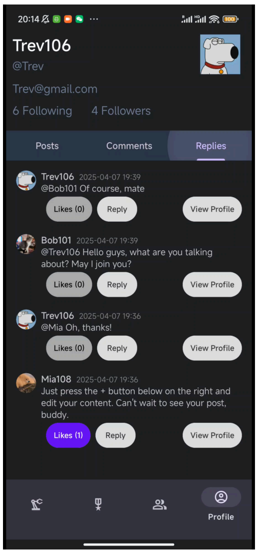
**Profile**
On the Profile page, users can review their **posts, comments,** and **replies** by navigating through the bar under their personal information view. They can also update their icons or change their usernames by clicking the corresponding areas. Users can change their avatar and user name by directly clicking the corresponding item. Moreover, users can check out their following and follower lists by directly clicking the “Following” and “Followers”.



Posts



Comments



Replies

# 2. Description of All Features

- **Course Management and Watch Record**
    - **Add Course**
      Clicking the **+** button on the Course Page takes users to the Add Course Page, where they can add a new course by simply pasting a YouTube link. The course information, including the title and thumbnail, will automatically display on the Course Page.
    - **Watch Youtube video and store the record**
      Clicking the **Study** button for a specific course navigates users to the Video Player Page. On this page, users can watch the video, and the app will automatically save their viewing time.
- **Ranking**

  There are three leaderboards: **Today**, **This Week**, and **This Month**, which show each user's ranking based on their study time during different periods. This helps motivate users to stay consistent. Users can also learn more about high-ranking individuals on the leaderboards by clicking on their icons to access their profiles.
- **Community**

- **View scrollable brief of each thread**
  - **View Scrollable Briefs of Each Thread**

    On the Community Page, users can scroll through a list of threads, each displaying a brief preview including the title, original post, and interactive Like and Comment buttons. The numbers next to each button are the respective counts of likes and comments.

  - **View detailed information of each thread**

    Clicking on a thread navigates users to the Thread Detail Page, where they can read the full post and all comments. Users can also share their thoughts by submitting new comments via the input box at the bottom of the page.

  - **Add New Posts and Comments**

    To encourage engagement, users can create new posts by clicking the + button located at the bottom right corner of the Community Page.

  - **View Users Profile Inside Posts**

    Additionally, users can visit the profile pages of anyone who posts or comments, allowing them to explore others' content and build connections.

of the page.

Additionally, you can visit profile of each user who posts the thread or comments under the original thread.

- **Watch Youtube video and store the record**
  Clicking the **Study** button for a specific course navigates users to the Video Player Page. On this page, users can watch the video, and the app will automatically save their viewing time.

- **Profiles**
  - **History of Posts, Comments, and Replies**
    Users can check out their posts and comments in the post section which ensures that users can easily navigate and review their content without unnecessary complexity.
  - **Following**
    To foster connections and create a sense of belonging, we let users follow each other freely. By following others, users can easily keep up with the latest posts

and comments from their favorite accounts, gaining deeper insights into each other’s interests and opinions. On their profile pages, the numbers of followers and following are displayed, giving users a sense of connection and fulfillment.

- **Personalization**
  Users have the freedom to customize their avatars and user IDs to reflect their individuality and preferences. This feature not only enhances personalization but also serves as a form of self-expression.

- **Account and Password**
  We have implemented an account and password system to provide users with a more secure and convenient experience. With this feature, users can easily log in, access their personalized content, and keep their data safe.

## 3. Top 3 Most Interesting/Technically Challenging Features

- **Network Connectivity and Asynchronous Operations**
  One of the most challenging features of our app is implementing robust network connections and handling asynchronous operations efficiently. Our app includes several interactive components, such as enabling users to create posts, comment on and reply to other users' posts, follow each other, and view each other's profiles. These functions require seamless real-time data synchronization across devices. To achieve this, all network operations are handled asynchronously using the **OkHttpClient**. This ensures the app remains responsive while waiting for server responses. The **enqueue** method is used to execute HTTP requests in the background, and callbacks are implemented to handle server responses or errors. For example, when creating a user, the app sends a request to the server and uses callback methods like **onResponse** and **onFailure** to process the server's response or handle failures.

- **Data Storage**
  Another challenging part is the data storage, because it requires reliable synchronization between the app and the remote database. User data, such as **userID, hashedPassword, username, email,** and **avatarVersion**, is converted into JSON objects and sent to a remote server. Our app uses HTTP POST requests to endpoints (e.g., /create_user, /update_user) to store or update user information in the remote database. Additional data, like posts, comments, and study records, is also stored through similar JSON-based API calls.

- **Cross-Platform Integration with Learning Tracking**

  The application is initially designed to learn courses. Youtube is one of the most popular video platform with an abundance of free video learning materials. To get the resources from Youtube platform. The technical implementation involves using YouTube's Data API v3 with an API key for authentication to fetch video metadata. The app extracts video IDs using regex pattern matching in the extractVideoId() method.

Video detail is retrieved by AsyncTask to prevent blocking, with the results parsed from JSON responses. It encoded video titles, URLs, and other metadata. The viewing experience is handled with library com.pierfrancescosoffritti.androidyoutubeplayer. It has a player that can track playback events. The app keeps track of time data with the backend database using OkHttp for REST API communication and stores timestamps and progress metrics that match the user's profile and contribute to community leaderboards.

# 4. Build Instructions

1. To install our app, you can either clone the repository from [https://github.com/TrevorChLiu/LearnLeague](https://github.com/TrevorChLiu/LearnLeague) or open our submitted zip file.
2. There is a folder named *LearnLeague*, which is the main part of our project. And there is a Python file *remote_server.py*, which is a Python server providing an API to manipulate data in the remote database.
3. Please download the required dependencies by following the ***README.md*** in our project (highly important!!).
4. Open the project in Android Studio and sync Gradle. Make sure your Android API version is **at least 33**.
5. Build the project and run the app on an emulator or physical device.

# 5. Reflection on the Project

- **Supported Online Platform**
  We wanted to support more online video platforms such as Bilibili and Khans Academy etc.., but due to the time constraint, we have only included YouTube as it is the largest video platform in the world and it can cater for more people from different nations.
- **Matching Mechanism**
  The original plan was to match users based on some interest tags to enhance relevance. However, due to time constraints, this feature has not been implemented. Future iterations will prioritize integrating this mechanism to improve user interactions.
- **Data Storage**
  Now our data is stored by MySQL in a remote server, which has led to longer load times and may affect user experience. In the next version, we should explore more efficient storage solutions, such as NoSQL databases or in-memory data stores, to optimize performance and scalability.